Zill's book : 4.6 P-146

## Lecture 09: Method of variation of parameter:

This method will be used for determining a particular solution $y_p$ of a non-homogeneous linear ODE that has, in theory, no such restrition on it. This method due to the eminent astronomer and mathematician Joseph Louis Lagrange (1736-1813), is known as "variation of parameters method."

**Wronskian:** Suppose each of the solution $y_1 = f_1(x)$, $y_2 = f_2(x)$ ---- $y_n = f_n(x)$ possesses at least $n-1$ derivatives. The determinant

$$W(y_1, y_2, \ldots y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \vdots & \vdots \\ y_1^{n-1} & y_2^{n-1} & & y_n^{n-1} \end{vmatrix}$$

is called the Wronskian of the solution. If $W \neq 0$, then the solutions are linearly independent and this method is applicable for determining $y_p$, that mean, if $W = 0$ then this method fails.

Linear 2nd order ODE: Suppose $a_2 y'' + a_1 y' + a_0 y = g(x)$

$$\Rightarrow \quad y'' + P y' + Q y = f(x) \quad \text{----} \quad (1)$$

A.E. $m^2 + pm + q = 0$

$\therefore \; m = m_1, m_2$

$$\boxed{y_c = c_1 y_1(x) + c_2 y_2(x)} \quad \text{----} \quad (*)$$

Let $\quad y_p = u_1(x) y_1 + u_2(x) y_2 \quad \text{----} \quad (2)$

$$y_p' = u_1 y_1' + \underline{u_1' y_1} + u_2 y_2' + \underline{u_2' y_2} \quad \text{----} \quad (3)$$

If we choose $u_1$ and $u_2$ such that

$$\boxed{u_1' y_1 + u_2' y_2 = 0} \quad \text{----} \quad (4)$$

(3) becomes, $\quad y_p' = u_1 y_1' + u_2 y_2'$

$$y_p'' = u_1 y_1'' + u_1' y_1' + u_2 y_2'' + u_2' y_2'$$

Substituting $y_p''$, $y_p'$ and $y_p$ into (1), we get

$$u_1 y_1'' + u_1' y_1' + u_2 y_2'' + u_2 y_2'' + u_2' y_2' + P(u_1 y_1' + u_2 y_2') + Q(u_1 y_1 + u_2 y_2) = f(x)$$

$$\Rightarrow \quad u_1 \underbrace{(y_1'' + P y_1' + Q y_1)}_{\to 0} + u_2 \underbrace{(y_2'' + P y_2' + Q y_2)}_{\to 0 \text{ (See *)}} + u_1' y_1' + u_2' y_2' = f(x)$$

$$\Rightarrow \quad u_1' y_1' + u_2' y_2' = f(x) \quad \text{----} \quad (5)$$

$$④ \Rightarrow \left.\begin{aligned} y_1 u_1' + y_2 u_2' &= 0 \\ y_1' u_1' + y_2' u_2' &= f(x) \end{aligned}\right\} \text{-----} (**)$$

By Cramer's rule, the solution of the system $(*,*)$ as

$$D = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \text{Wronskian} = W \neq 0$$

$$D_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix} = -y_2 f(x)$$

$$D_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix} = y_1 f(x)$$

$$\therefore \quad u_1' = \frac{D_1}{W} = \frac{-y_2 f(x)}{W}$$

$$u_2' = \frac{D_2}{W} = \frac{y_1 f(x)}{W}$$

$$\therefore \boxed{\begin{aligned} u_1 &= \int \frac{-y_2 f(x)}{W}\, dx \\ u_2 &= \int \frac{y_1 f(x)}{W}\, dx \end{aligned}}$$

Example 01: Solve $y'' - 4y' + 4y = (x+1)e^{2x}$

Ans: A.E, $m^2 - 4m + 4 = 0$

$$m = 2, 2$$

$$y_c = c_1 e^{2x} + c_2 x e^{2x}$$

$$W = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & 2xe^{2x}+e^{2x} \end{vmatrix} = \cancel{2xe^{4x}} + e^{4x} - \cancel{2xe^{4x}}$$

$$= e^{4x} \neq 0$$

$$u_1 = -\int \frac{y_2 f(x)}{W}\,dx = \int \frac{x\cancel{e^{2x}}(x+1)\cancel{e^{2x}}}{\cancel{e^{4x}}}\,dx$$

$$u_1 = \int (x^2+x)\,dx = \frac{x^3}{3} + \frac{x^2}{2}$$

$$u_2 = \int \frac{y_1 f(x)}{W}\,dx = \int \frac{\cancel{e^{2x}}(x+1)\cancel{e^{2x}}}{\cancel{e^{4x}}}\,dx = \int (x+1)\,dx$$

$$u_2 = \frac{x^2}{2} + x$$

$$\therefore\ y_p = u_1 y_1 + u_2 y_2 = \left(\frac{x^3}{3}+\frac{x^2}{2}\right)e^{2x} + \left(\frac{x^2}{2}+x\right)xe^{2x}$$

$\therefore$ G.S. $y = y_c + y_p$

<u>Example 02:</u> Solve $4y'' + 36y = \operatorname{cosec} 3x$

<u>A:</u> Given $4y''+36y = \operatorname{cosec} 3x$

$$\Rightarrow\ y'' + 9y = \frac{1}{4}\operatorname{cosec} 3x \quad \text{---} (1)$$

A.E; $m^2 + 9 = 0 \Rightarrow m = \mp 3i$

$y_c = A\cos 3x + B\sin 3x$

$$W = \begin{vmatrix} \cos 3x & \sin 3x \\ -3\sin 3x & 3\cos 3x \end{vmatrix} = 3\cos^2 3x + 3\sin^2 3x = 3$$

$$u_1' = -\int \frac{y_2 f(x)}{W}\,dx = -\int \frac{\sin 3x \,\frac{1}{4}\csc 3x}{3}\,dx = -\int \frac{1}{12}\,dx$$

$$u_1 = -\frac{1}{12}x$$

$$u_2 = \int \frac{y_1 f(x)}{W}\,dx = \int \frac{\cos 3x \,\frac{1}{4}\csc 3x}{3}\,dx = \frac{1}{12}\int \frac{\cos 3x}{\sin 3x}\,dx$$

$$= \frac{1}{3\times 12}\ln|\sin 3x|$$

$$y_p = u_1 y_1 + u_2 u_2 = -\frac{1}{12}x\cos 3x + \frac{1}{36}\ln|\sin 3x|\sin 3x$$

G.S. $y = y_c + y_p$.

Example 03: Solve $y'' + y = \tan x$

Ans: We will solve in the class

Home task: Exercise 4.6 Page-161

4, 8, 9, 11–18

Assignment 02: If $a_3 y''' + a_2 y'' + a_1 y' + a_0 y = g(x)$

Find $u_1'$, $u_2'$ and $u_3'$.

Use above formula and solve the following ODEs: 25, 26, 27, 28

Due date: 26/08/20